MedTaxon: Accelerating Biomedical Taxonomy Construction with A Web-based Interactive Tool using Large Language Models

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Introduction

In the realm of biomedical research, taxonomies can provide hierarchical classifications of concepts within a specific domain. They serve as essential frameworks for structuring vast biomedical information, which enable researchers to navigate through complex datasets, identify relevant information, and establish connections between disparate pieces of knowledge[1]. Traditionally, the construction of biomedical taxonomies has mainly relied on manual curation by domain experts, which is inherently a time-consuming, labor-intensive, and costly endeavor. Although some automated approaches offer the potential for handling large volumes of data more efficiently, they may also struggle with the identification of synonyms and often require significant post-processing refinement by human experts to ensure semantic accuracy and comprehensive coverage within a domain[2].

The emergence of large language models (LLMs) with their advanced natural language understanding and generation capabilities, has demonstrated the potential to automate or at least semi-automate various knowledge organization processes with minimal human intervention while achieving high levels of performance[2]. Therefore, to address the challenges in constructing a taxonomy from textual knowledge, we propose MedTaxon, a serverless web-based tool to assist researchers in developing a domain taxonomy. MedTaxon integrates interactive user interfaces (UIs) and LLMs to assist researchers in extracting and structuring domain-specific taxonomies, aiming to reduce the manual effort and improve the overall efficiency.

Method

The overall workflow of biomedical taxonomy creation, which typically includes many steps, such as resource integration, taxonomy prototyping, dataset generation, iterative taxonomy dataset development, and evaluation[1]. Among these steps, ***iterative taxonomy dataset development***—the step involving the annotation of relevant biomedical literature, agreement evaluation, and consensus development—is usually the most time-consuming, requiring extensive manual effort to ensure taxonomy's comprehensiveness and accuracy. To understand the challenge of this step, we conducted several rounds of interviews with biomedical researchers experienced in ontology development and taxonomy development with initial conceptual prototype designs. With their input, we identified that the key challenge in this step is the ***fragmented workflows and frequent context-switching*** which lead to cognitive overload.

For instance, to annotate a document based on a taxonomy, researchers typically need to simultaneously handle multiple files and software applications for diverse information sources, such as compiling *dataset file* in Excel, examining similar cases and cross-referencing a *taxonomy file* in Word, and reviewing annotation guideline PDF file in a PDF viewer. Meanwhile, each annotation may demand vastly different domain-specific knowledge which needs to be searched online in a web browser. This fragmented process leads to significant mental effort as researchers repeatedly shift cognitive contexts, and results in substantial inefficiencies, slowing progress, increasing errors, and elevating the cognitive burden associated with building accurate, comprehensive biomedical taxonomies.

Integrating multiple LLMs into the taxonomy creation workflow

MedTaxon offers a solution by integrating multiple LLMs and interactive UIs to optimize the workflow. It automates document annotation, minimizing repetitive manual tasks by consolidating literature metadata and LLM-generated suggestions within a single interface. Researchers can then iteratively build the corpus by reviewing these suggestions through intuitive interfaces. This human-in-the-loop methodology significantly expedites corpus creation, empowering researchers to efficiently develop accurate, domain-specific taxonomies.

The proposed MedTaxon system, as illustrated in Figure 1(a), begins with the user uploading taxonomy, dataset, and prompt template files using the provided editing interfaces. Each file is just a plain text file or tab-separated values (TSV) file, which used to be opened by different softwares. For each document in the dataset file, prompts are generated based on the taxonomy and prompt template, which are subsequently classified by LLMs. The LLMs generate multiple category suggestions with explanations. These suggestions are then manually reviewed by the user for making a final decision. The user can also refine both the taxonomy and the prompt template. Additionally, the refined taxonomy structure and dataset can be visualized, allowing for clear tracking of taxonomy distribution. Finally, through this iterative human-in-the-loop process, all the documents in the dataset file can be efficiently processed and saved locally. Users can access and modify it using external text editors.

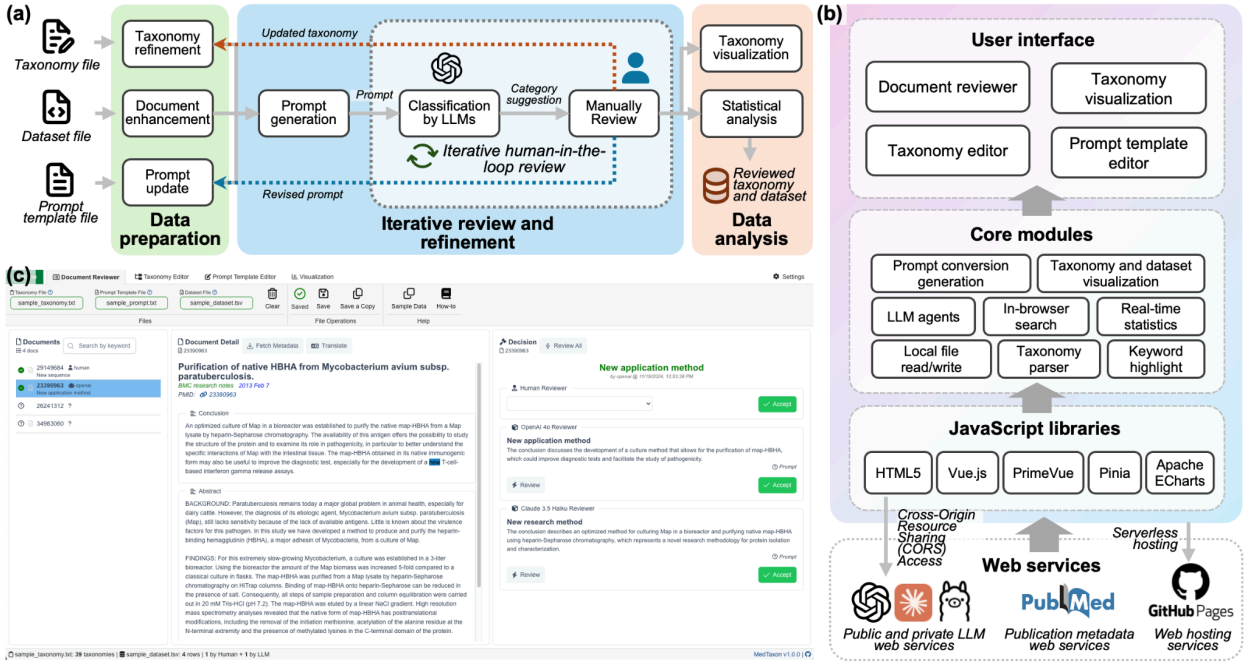

Figure 1: (a) The overall workflow based on the proposed system. (b) The architecture of the proposed MedTaxon system, including user interface, core modules and JavaScript libraries. (c) The screenshot of MedTaxon.

System architecture

To simplify the use of MedTaxon for researchers, we adopted a serverless architecture to design the system, which eliminates the need for a traditional always-on server and ensures storing all data files locally. MedTaxon offers specialized UIs, including a document reviewer, taxonomy editor, taxonomy visualization interface, and prompt template editor, to support the iterative development of taxonomies, particularly the annotation of biomedical literature. Built with modern JavaScript libraries, MedTaxon provides functionalities such as prompt generation, in-browser search, real-time statistics, and keyword highlighting within a unified interface. Importantly, MedTaxon integrates multiple LLM agents through public and private web services (e.g., OpenAI, Anthropic, or other OpenAI-compatible APIs). This eliminates the need for users to install additional LLM software or manage dedicated servers. Users can use LLM agents by configuring API keys in their local web browser to use LLMs.

User interface

Figure 1(c) illustrates the UI of the document reviewer, which utilizes a three-column layout for organizing information used in the reviewing process. The left column displays a list of documents awaiting classification, along with their current classification status. The central column provides detailed views of the selected document, highlighting key terms to facilitate rapid understanding and contextualization. The right column showcases taxonomy category suggestions generated by LLMs, accompanied by explanations clarifying the rationale behind each suggestion. Users can accept these suggestions or specify the category manually. If needed, users can also refine the underlying prompts and taxonomy definitions. All information generated during the review process, such as LLM responses and user decisions, can be saved back to the original files on the user's local disk.

Discussion and Future Work

We are developing MedTaxon in conjunction with an ongoing taxonomy creation project. Domain experts have provided feedback indicating that our system increases efficiency and reduces time spent switching between different software, thereby aiding in large corpus creation. Both the system and sample data are currently available for demonstration at https://clinicalnlp.org/medtaxon/. Our next steps include integrating additional visualization functions to assist users with taxonomy analysis, as well as conducting case studies and formal usability evaluation.

Reference

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